DC MOTOR SPEED CONTROL USING PID CONTROLLER

Technical Project Report

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Abstract

This project studies closed-loop speed control of a DC motor using a PID controller in MATLAB/Simulink. Two simulation models are included: a reference-tracking model with a 1000 RPM target and a model intended to examine the effect of a load-torque disturbance. The motor speed is measured and fed back to the controller so that the difference between the reference and measured speed can be reduced. The report describes the system, operating principle, simulation workflow, recorded performance measures, and possible future improvements.

1. Introduction

DC motors are used in industrial drives, robotics, electric vehicles, and automation. Their speed can change when the mechanical load changes. A closed-loop control system measures the motor speed, compares it with a desired reference, and uses the resulting error to adjust the motor input. PID control combines proportional, integral, and derivative actions to regulate the response.

2. Objectives

- Develop and simulate a DC motor speed-control system in MATLAB/Simulink.
- Use a PID controller to track a 1000 RPM reference.
- Observe the response to a change in mechanical load using a second model.
- Calculate rise time, settling time, overshoot, final speed, and steady-state error from simulation data.
- Organize the plots and MATLAB scripts for project documentation.

3. System Description

The supplied models include a DC Motor, PID Controller, reference Step block, Sum block for calculating the speed error, Ideal Rotational Motion Sensor, Simulink-PS and PS-Simulink Converters, and Scope/display blocks. The load-disturbance model additionally includes an Ideal Torque Source and a step input used to apply the disturbance. The exact motor parameters, PID gains, load magnitude, and solver configuration should be read from the model files when documenting a final experiment.

4. Working Principle

The reference speed is compared with the measured shaft speed. The difference is the control error. The PID controller processes this error and produces a control command that drives the motor through the controlled voltage source. The rotational motion sensor measures the shaft motion and returns feedback to the summing junction. When a load torque is applied, the motor speed may deviate from the reference; the controller then adjusts its command to reduce the error.

$$u(t) = K_p e(t) + K_i \int e(t)\,dt + K_d \frac{de(t)}{dt}$$

where e(t) is the speed error, and Kp, Ki, and Kd are the proportional, integral, and derivative gains, respectively.

5. Simulation Procedure

1. Open DC_Motor_PID_1000rpm.slx or DC_Load_disturbance.slx in MATLAB/Simulink.
2. Check the reference input, PID gains, motor parameters, load-torque step, solver settings, and stop time.
3. Run the reference-tracking model and inspect the motor-speed output in the Scope.
4. Run the load-disturbance model and observe the speed before, during, and after the applied load.
5. Log the speed signal or export it to the MATLAB workspace.
6. Use the recorded time and speed arrays to calculate time-domain performance measures and create plots.

6. Results and Discussion

The supplied results PDF includes a reference-tracking plot and a load-disturbance response plot. The reference-tracking plot shows the output rising to the 1000 reference and then remaining close to it over the displayed simulation interval. In the supplied load-disturbance plot, the speed appears close to the reference at the displayed scale; a short or small transient may not be distinguishable in that figure.

Figure 1. Reference speed and actual motor-speed response

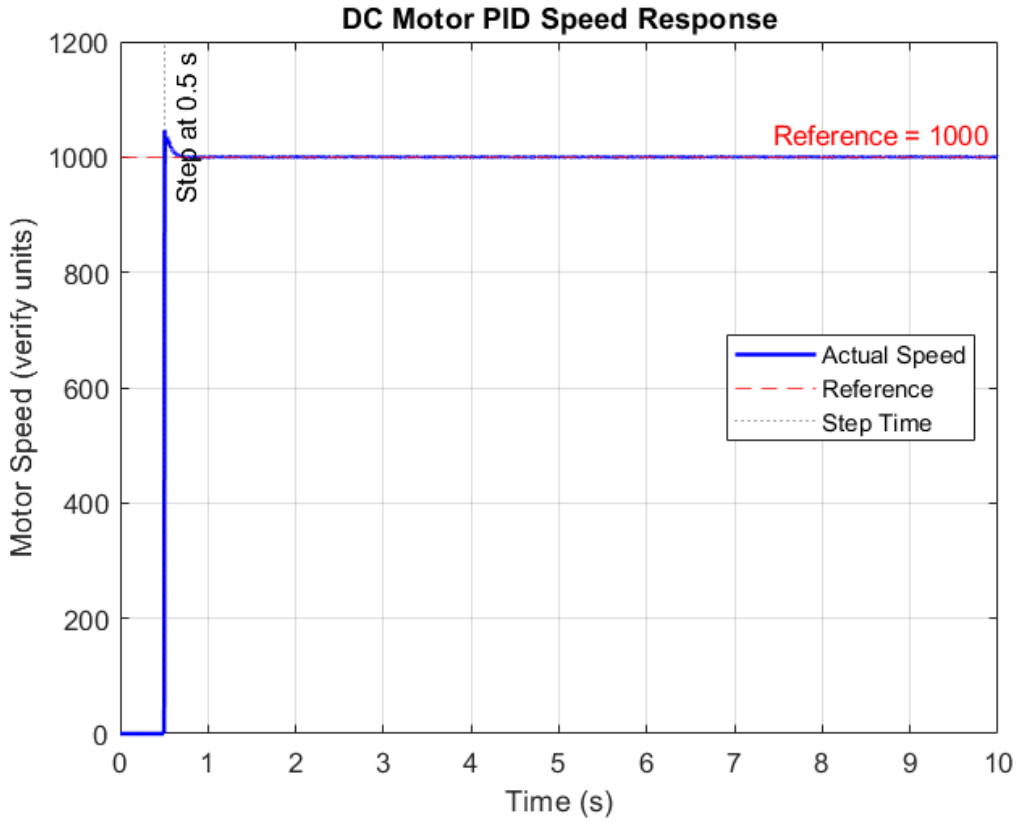
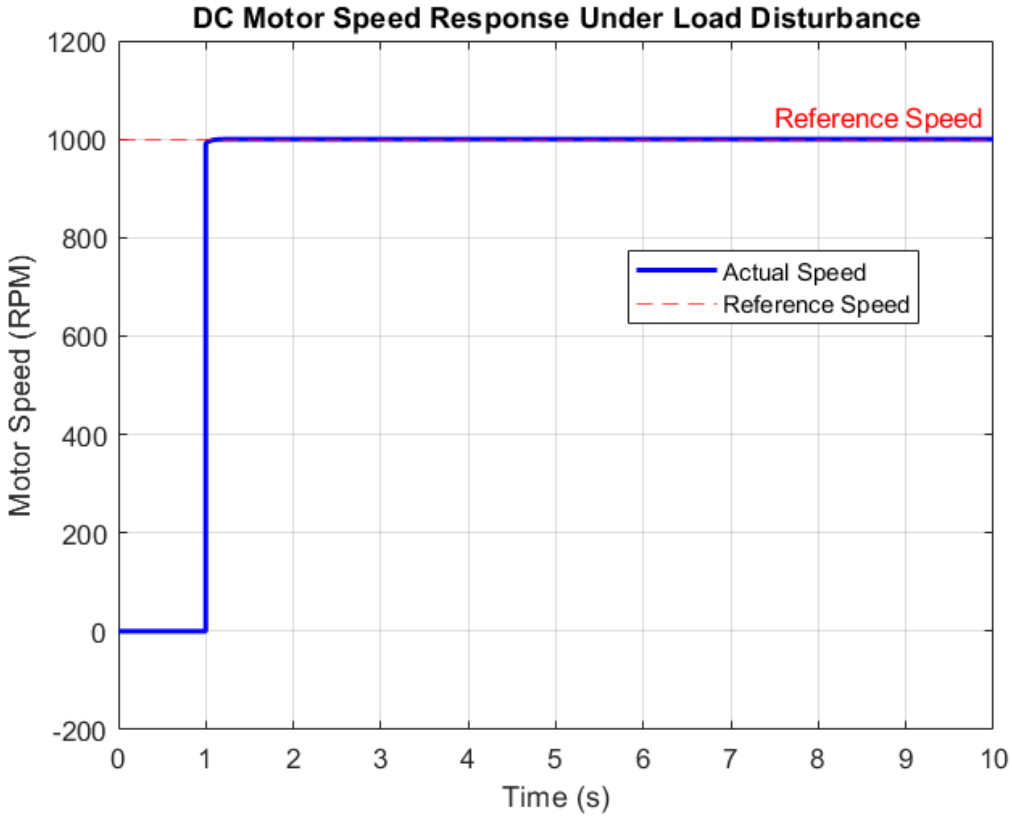


Figure 2. Motor-speed response under load disturbance (from supplied results PDF)



6.1 Performance Metrics

The later MATLAB Command Window output supplied during the analysis reported the following values for the reference-tracking model. The speed units should be confirmed from the model's sensor and conversion configuration before the results are labelled definitively as RPM.

Performance parameter	Recorded value
Reference speed	1000 (units to confirm; intended reference is 1000 RPM)
Rise time	0.004368 s
Settling time	0.069864 s
Percentage overshoot	4.6788%
Final speed	1000.0005 (signal units)
Steady-state error	−0.0005 (signal units; reference − final speed)

6.2 Note on Source Result Consistency

The supplied results PDF also contains earlier annotations in its transient-response and performance-metrics figures that differ from the later MATLAB Command Window output above. This report uses the later detailed console output for the numerical table. Before formal submission, regenerate the plotted metric annotations from the same simulation run so that all figures and tables report one consistent set of values.

7. Conclusion

The project demonstrates a Simulink workflow for PID-based DC motor speed regulation and provides a second model for examining load-torque disturbance effects. The recorded reference-tracking results show the output approaching the 1000 reference with a small final error. Quantitative conclusions about the load-disturbance response require inspecting the actual logged disturbance signal and its transient details. Consistent plots and performance values should be generated from the same simulation run before final submission.

8. Future Scope

- Tune the PID gains using a documented method such as Simulink PID Tuner and compare the results.
- Compare PID control with PI, fuzzy logic, or another suitable controller.

- Test different reference speeds and disturbance magnitudes.
- Evaluate robustness to changes in motor parameters and solver settings.
- Report disturbance speed dip and recovery time using logged data from the load-disturbance model.

References :

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